

2	(a) (i) sum of kinetic and potential energy of atoms/molecules reference to random (distribution)	M1	A1	[2]
	(ii) no forces (of attraction or repulsion) between molecules	B1		[1]
	(b) $pV = NkT$ or $pV = nRT$ and $R = kN_A$, $n = N/N_A$ $\frac{1}{3} Nm\langle c^2 \rangle = NkT$ or $\frac{1}{3} m\langle c^2 \rangle = kT$ $\langle E_K \rangle = \frac{1}{2} m\langle c^2 \rangle$ <u>so</u> $\langle E_K \rangle = \frac{3}{2} kT$	B1	B1	B1
				[3]
	(c) (i) $\langle E_K \rangle = \frac{3}{2} \times 1.38 \times 10^{-23} \times (273 + 12)$ $= 5.9 (5.90) \times 10^{-21} \text{ J}$	C1	A1	[2]
	(use of $T = 12 \text{ K}$ not $T = 285 \text{ K}$ scores 0/2)			
	(ii) number = $(17/32) \times 6.02 \times 10^{23}$ $= 3.2 (3.20) \times 10^{23}$	C1	A1	[2]

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(iii) internal energy = $5.9 \times 10^{-21} \times 3.2 \times 10^{23}$
= 1900 (1890) J

A1 [1]